

Supplementary Material

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Abstract

In the supplementary material, we include background information of the data set, method details of the reference panel construction, as well as data filtering strategies, and rationale behind the choice of parameters.

Contents

S1 Data and reference panel	S1-2
S1.1 Pf3k data set	S1-2
S1.2 Generating <i>in silico</i> mixtures for deconvolution benchmark	S1-2
S1.3 <i>In silico</i> mixtures of lab strains	S1-2
S1.4 Data quality	S1-2
S1.5 Variant selection	S1-2
S1.6 Variant quality score log-odds	S1-3
S1.7 Data cleaning and filtering	S1-3
S1.7.1 High leverage data points	S1-3
S1.8 Reference panel	S1-3
S2 DEploid-LASSO deep dive	S2-1
S2.1 Model	S2-1
S2.2 Effective reference panel	S2-1
S3 Major improvement from DEploid and DEploid-IBD	S3-1
S3.1 Optimised reference panel per sample	S3-1
S3.2 Reduced run time	S3-1
S3.3 Improved inference for balanced mixtures	S3-1
S3.4 Improved inference results for mixtures of 3- or 4-strains	S3-1
S4 Github actions	S4-1
S5 Modularized package development	S5-1
S5.1	S5-1
S6 Deconvolution of the field samples	S6-1

S1 Data and reference panel

S1.1 Pf3k data set

The Pf3k (Pf3k Consortium, 2016) dataset includes 2,640 *Plasmodium falciparum*

can be divided into field and lab samples based on their sources, and clonal and mixed samples based on the mixtures based on our work (Zhu et al., 2018, 2019), see Table S1.1 for details. Note, we drop the duplicated samples and use only the preferred ones.

TODO: check We also exclude samples where sequencing depth is below 30 and less than 30% of sites are callable.

	Clonal samples	Mixed samples	Total
Field samples	1365	979	
Lab samples			
Total			

Table S1.1: Field samples are collected from multiple sampling sites in Africa and Asia (Duffy et al., 2015; Kamau et al., 2015; MalariaGEN *Plasmodium falciparum* Community Project, 2016; Mobegi et al., 2014; Amato et al., 2017; Miotto et al., 2013; Ocholla et al., 2014; Amambua-Ngwa et al., 2012; Wong et al., 2017). Clonal and mixed field samples are identified by Zhu et al. (2019). The lab clonal samples include 5 lab clonal samples (7G8, GB4, KH02, KE01, GN01) and 96 [check this number](#) crosses samples comprising parents and progeny from the *P. falciparum* Genetic Crosses project (Miles et al., 2016). The lab mixed samples include 27 mixed lab strains in varying proportions created by Dr Jason Wendler (Wendler, 2015).

S1.2 Generating *in silico* mixtures for deconvolution benchmark

S1.3 *In silico* mixtures of lab strains

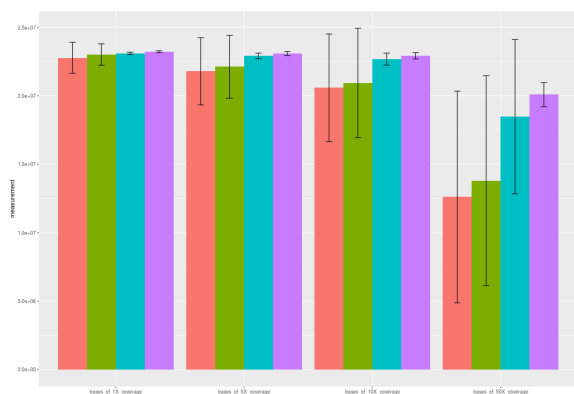
In addition to the *in vitro* lab mixtures from Wendler (2015), we generated four *in silico* lab mixtures: We used the lab strain GB4 to provide a whole-genome read depth profile. We drew Bernoulli variables to simulate alternative allele counts, with the probability of success as the adjusted within sample allele frequency at each position. The expected WSAF is the dot product of the genotype of the lab strains: 3D7, Dd2, HB3, 7G8 and the proportions. For example, if the genotype is 0, 0, 1, 1 at an arbitrary position, and the mixture proportion of 0.1, 0.2, 0.4 and 0.3 will lead to an expected WSAF of 0.7. We then further adjust the WSAF with an error term (0.01) to account for all technical error/noise: $0.7 \times (1 - 0.01) + (1 - 0.7) \times .01 = 0.696$.

S1.4 Data quality

S1.5 Variant selection

We ran DEploid-IBD on high quality biallelic SNPs (both coding and non-coding, tagged with PASS at the QUAL column in the VCF file) from Pf3k (Pf3k Consortium, 2016). Before the additional filtering described below, this set contained 1,057,830 SNPs.

```
bcftools view \
--include 'FILTER="PASS"' \
--min-alleles 2 \
--max-alleles 2 \
--types snps \
--output-file SNP_INDEL_Pf3D7_01_v3.high_quality_biallelic_snps.vcf.gz \
--output-type z \
SNP_INDEL_Pf3D7_01_v3.combined.filtered.vcf.gz
```

A coverage**B VQLSOD**

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S1.6 Variant quality score log-odds

The variant quality score log-odds (VQSLOD) score measures how reliable each variant call is. For the Pf3k data set, all ‘PASS’ sites have VQSLOD score above zero.

get a table

chromosomes, by row,, col

S1.7 Data cleaning and filtering

S1.7.1 High leverage data points

Same as Zhu et al. (2018, 2019), we find markers with high frequencies in both reference and alternative allele count can mislead our model to fit extra strains. We used a threshold of $\geq 99.5\%$ coverage (default) to identify markers with extremely high allele counts. We further expand this list of potential “bad markers” by considering their nearest 10 neighbours on both sides along the genome, and identify the nearest markers as “to be excluded” if overlaps are found (see Figure S1.1). These poorly-genotyped variants are likely to be artefacts of mapping and genotype calling.

This process remains as a manual step, as it may consider other sites.

S1.8 Reference panel

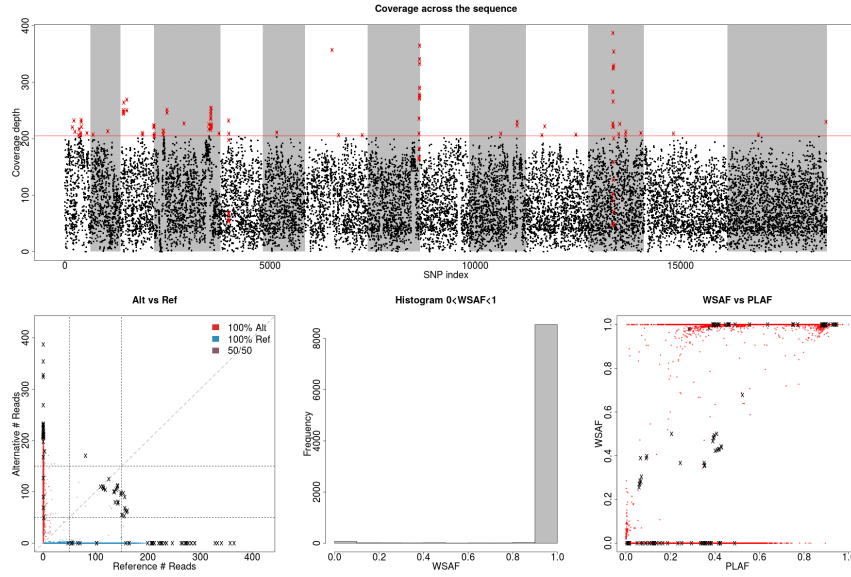


Figure S1.1: Figure on the top plot total allele counts across all markers. We observe a small number of heterozygous sites with high coverage (marked as crosses above), which can potentially mislead our model to over-fit the data with additional strains. We use a threshold of $\geq 99.5\%$ coverage to identify markers with extremely high allele counts. Red crosses indicate markers that are filtered out. Figure on the left plots the alternative allele count against the reference allele count. As *P. falciparum* genomes are haploid, in clonal samples, one would expect to see either alternative or reference allele at any sites. Heterozygous sites are indications of mixed infection. The marked black crosses can mislead our model to overestimate the number of strains as two. Figure in the middle is the histogram of the allele frequency within sample. Figure on the right show allele frequency within sample, compare against the population average.



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Figure S1.2: PCA/tSNE/UMAP of the big panel

S2 DEploid-LASSO deep dive

S2.1 Model

Assume Independence

S2.2 Effective reference panel



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Figure S3.3: bar plot of time, four bars, no panel, with panel, ibd, best practice

S3 Major improvement from DEploid and DEploid-IBD

S3.1 Optimised reference panel per sample

S3.2 Reduced run time

S3.3 Improved inference for balanced mixtures

S3.4 Improved inference results for mixtures of 3- or 4-strains

S4 Github actions

S5 Modularized package development

To facilitate the development and maintenance of the DEploid software, we have modularized the package into several components. Each component is responsible for a specific functionality, making the codebase more organized and easier to manage. The main modules are:

- **DEploid-Lasso-lib (C++)**: Handles the input and preprocessing of genomic data.
- **DEploid-vcf-lib (C++)**: Contains the core algorithms for deconvolution and inference.
- **DEploid.utils (R)**: Utility functions for data preprocessing,
- **DEploid (C++)**: DEploid core functions, command line interface.
- **DEploid-r (R)**: An R interface for DEploid, allows user to directly interact with DEploid results, without save the results in text files and load them again for analysis. Upstream direct dependencies include DEploid, DEploid.utils.
- **DEploid-py (Python)**: A python interface for DEploid, allows user to directly interact with DEploid results, without save the results in text files and load them again for analysis.

This modular approach allows for independent development and testing of each component, improving the overall robustness and flexibility of the software.

S5.1

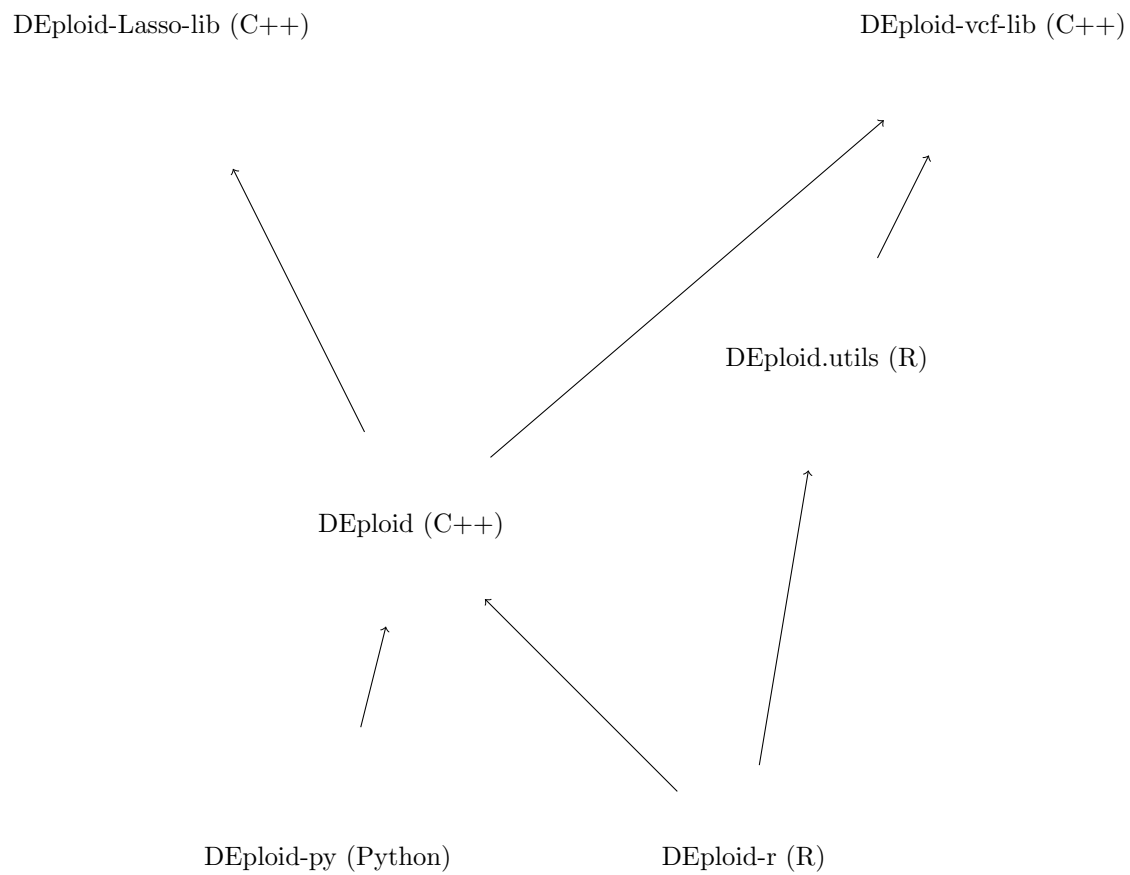


Figure S5.4: Package dependencies.

S6 Deconvolution of the field samples

Different than before, we are going for whole genome



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Figure S6.5: PCA

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